LangSmith

Presented by

Valle Ramesh



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The Need for LangSmith

Challenges with LLM Applications:

- LLM-based applications can produce **unexpected or incorrect results**.
- Hallucinations
- Inconsistent responses
- Quality issues

What is LangSmith?

- Lang Smith is a platform for **Tracing, evaluating, and improving applications** powered by Large Language Models (LLMs).
- Used by **AI/ML Developers, LLM Application Developers, Engineers building chatbots**, agents and AI applications

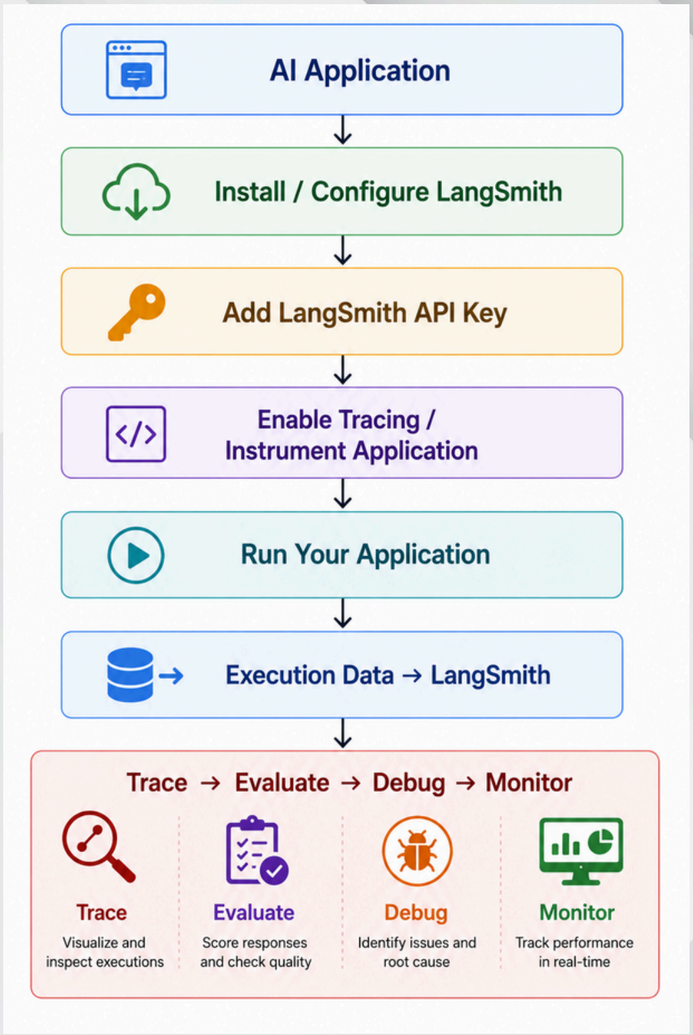
How Lang Smith Works

Install Python library (SDK)

- pip install langsmith

configuration conceptually:

- LANGSMITH_TRACING=true
- LANGSMITH_API_KEY=your_api_key

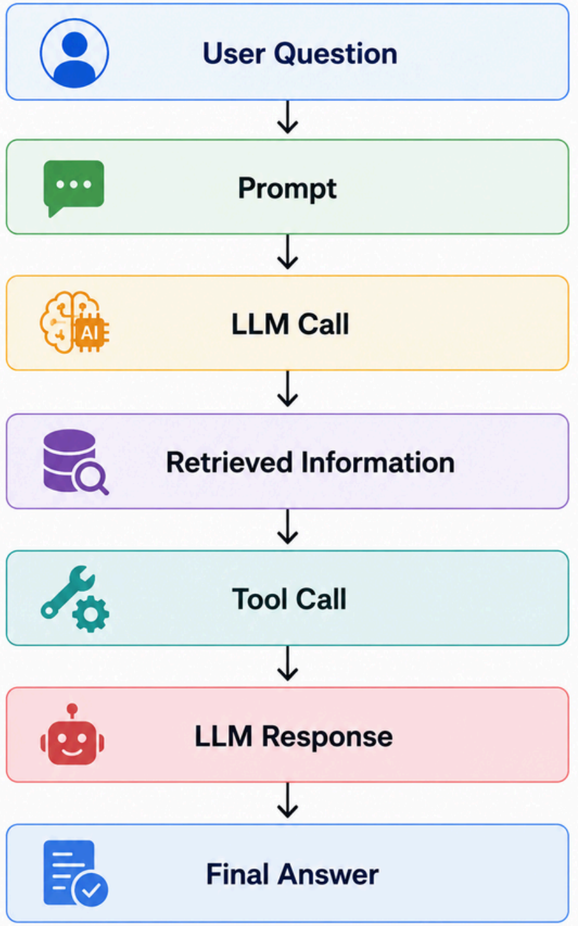


Tracing

- Tracing records the execution of an AI application's individual steps.

Tracing helps developers:

- Answer **work flow**
- Inspect **inputs** and **outputs**
- Follow **multi-step AI workflows**
- Identify **failed steps** & **unexpected responses**

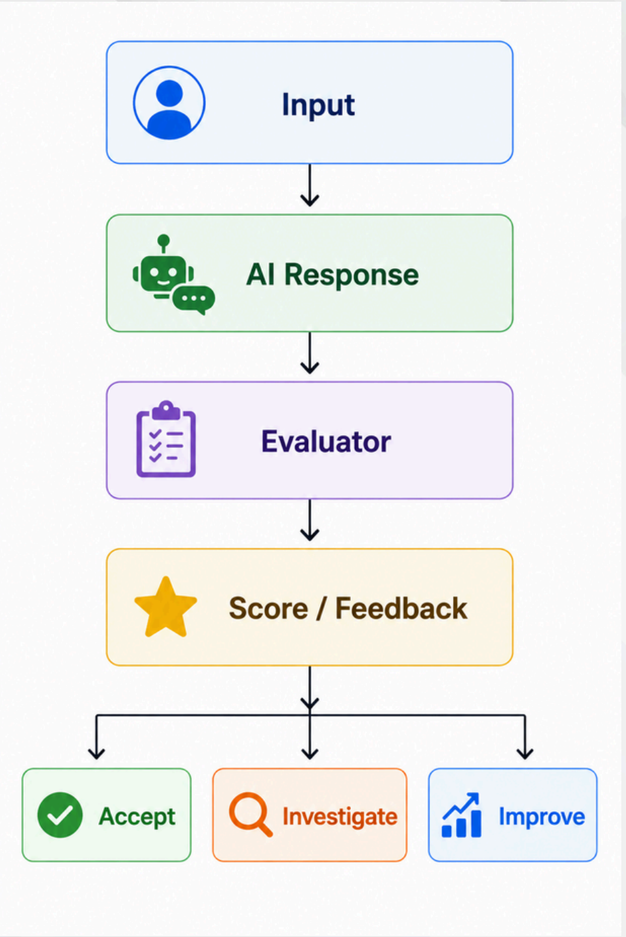


Evaluation

Evaluation measures how well an AI application performs.

Possible evaluation dimensions:

- **Correctness**
- **Relevance**
- **Quality**
- **Safety**
- **Reliability**



Types of Evaluation

Automated Evaluation

- Evaluation performed programmatically using defined criteria or evaluators.

Human Evaluation

- Human reviewers assess AI responses and provide feedback.

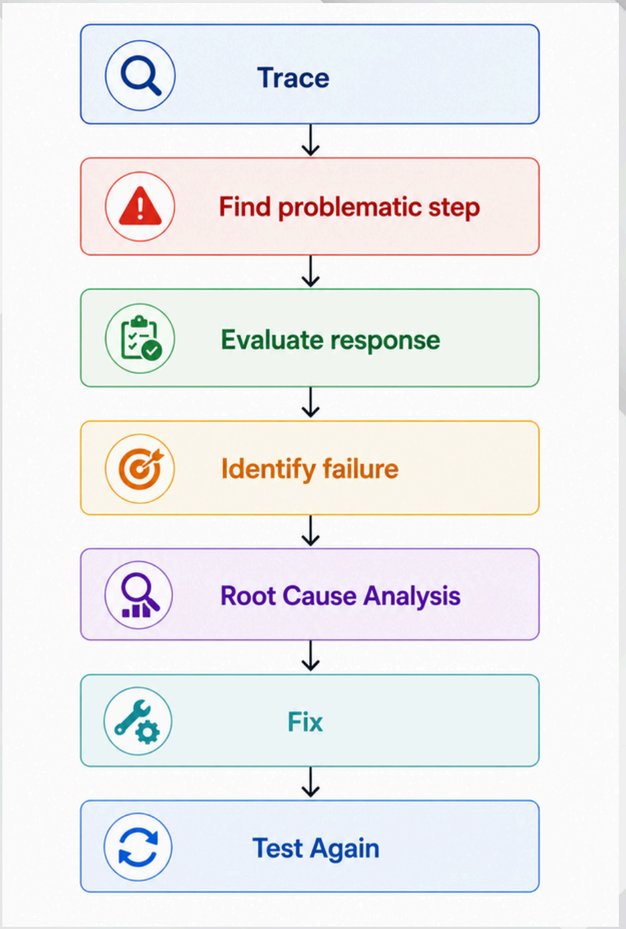
LLM-as-a-Judge

- Another LLM evaluates the quality of an AI-generated response according to defined criteria.

Debugging

Possible causes:

- Poor prompt
- Wrong retrieved information
- Tool failure
- Model limitation
- Incorrect application logic
- Missing context



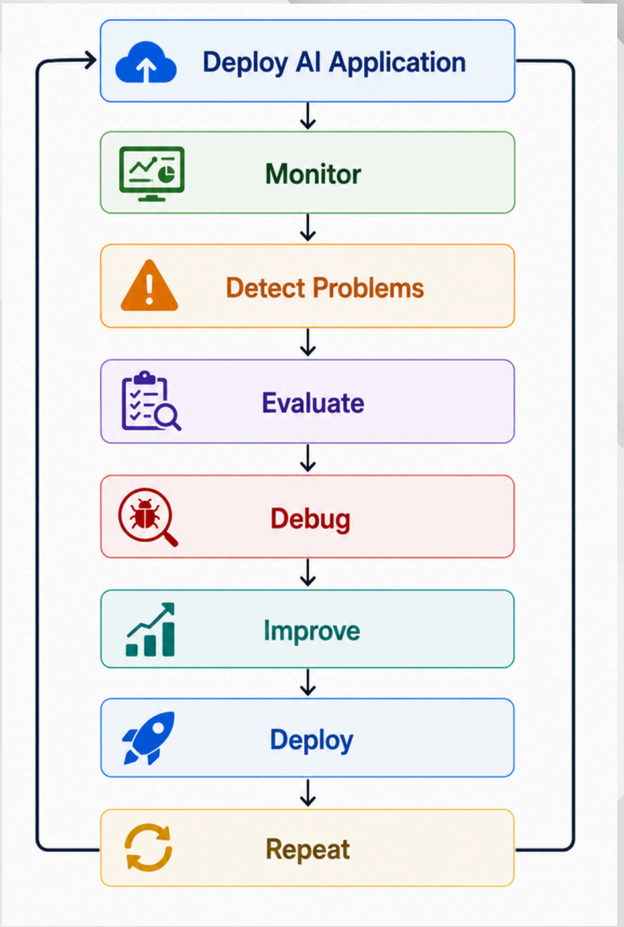
Monitoring & Continuous Improvement

Debugging

- Investigating a particular problem.

Monitoring

- Observing application behavior over time.



Real-World Case Study: CampusBot

Student:

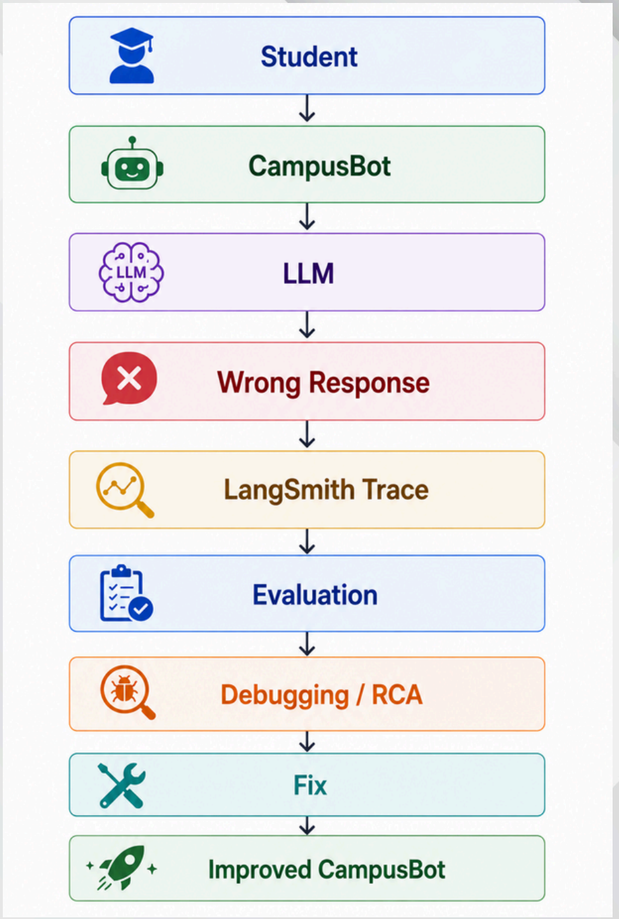
- What is the deadline for semester fee payment?

Campus Bot:

- August 15.

Actual university deadline:

- August 25 (Wrong Answer)



Benefits, Limitations

Benefits

- Debugging
- Evaluation
- Continuous improvement
- Complex LLM applications

Limitations

- Setup is difficult
- AI evaluation
- Doesn't automatically solve
- Human judgment necessary.

Final takeaway

LangSmith helps transform an LLM application from a system that simply generates responses into a system that can be observed, evaluated, debugged and continuously improved.

Any Queries?

Thank You

